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International Journal of Methods in Psychiatric Research

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ORIGINAL ARTICLE OPEN ACCESS

Psychometric Assessment of the Arabic Translation of the Adult ADHD Self-Report Scale Version 1.1 6-Question Screener

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Received: 27 January 2026 | **Revised:** 3 May 2026 | **Accepted:** 18 May 2026

Keywords: adult ADHD | Arabic | ASRS | Lebanon | psychometric validation | screener

ABSTRACT

Background: In the Middle East and North Africa, there is a dearth of research on Adult Attention-Deficit/Hyperactivity Disorder (ADHD). Offering a brief, publicly available, and psychometrically valid screening instrument for adult ADHD that can be easily used in large-scale surveys among Arabic-speaking populations could encourage research and facilitate case detection. The purpose of this study was to translate the Adult ADHD Self-Report Scale v1.1 6-Question Screener (ASRS-6) into Arabic and assess its psychometric properties among a sample of Lebanese adults from the general population.

Methods: A cross-sectional study was conducted between April and May 2025 using an online survey distributed via a mixed convenience and snowball sampling. A total of 1059 Lebanese adults participated (mean age = 28.52 ± 11.41 years; 66.0% females). The questionnaire included the Arabic ASRS-6, the validated Arabic versions of the Stress Numerical Rating Scale-11, the Patient Health Questionnaire, and the World Health Organization-Five Well-Being Index. The total sample was randomly divided into two split-half subsamples, one for the exploratory (EFA) and the second for the confirmatory (CFA) factor analysis.

Results: The EFA and CFA supported the one factor structure, which showed acceptable fit indices. A correlation was added between residuals of items 5 and 6, which improved the model fit and was consequently used in the analysis. A Cronbach's alpha of 0.68 indicated acceptable internal consistency. A significantly higher ASRS-6 mean scores was found in females versus males (10.85 ± 3.98 vs 10.22 ± 4.09; t (1057) = −2.41, p = 0.009, Cohen's d = 0.157). A significantly higher ASRS-6 mean score was found in single participants compared to married ones (10.84 ± 3.95 vs 10.19 ± 4.16; t (1057) = 2.46, p = 0.014, Cohen's d = 0.163). However, no significant difference was found between participants with a secondary level of education or less compared to those with a university education level in terms of ASRS-6 scores (10.20 ± 4.32 vs 10.73 ± 3.95; t (1057) = −1.68, p = 0.094, Cohen's d = 0.131).

Sahar Obeid, Feten Fekih-Romdhane, and Souheil Hallit are last coauthors.

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Higher ASRS-6 scores were significantly associated with higher depression ($r = 0.33$; $p < 0.001$), anxiety ($r = 0.33$; $p < 0.001$), stress ($r = 0.30$; $p < 0.001$), and lower well-being scores ($r = -0.38$; $p < 0.001$).

Conclusion: The results provide initial support for applying the Arabic version of the ASRS-6 screener in community epidemiological surveys, clinical outreach, and case-finding initiatives. However, additional work is needed to approve its validity and expand its use in different Arab nations.

1 | Introduction

Attention-Deficit/Hyperactivity Disorder (ADHD) characterized by disruptive symptoms of inattention and/or hyperactivity-impulsivity, is a complex neurodevelopmental disorder with a multifactorial etiology involving genetic and environmental factors (Assaf et al. 2024). It is typically identified in early school years and one of the most prevalent childhood disorders, but it may also endure into adolescence and adulthood. The global prevalence of ADHD in children is estimated at 5%, while persistent adult ADHD (with a childhood onset) is around 2.58%, and symptomatic adult ADHD (regardless of its onset during childhood) is 6.76%, signifying important public health burden (Song et al. 2021). Depending on the predominant symptoms, ADHD is divided into three presentations: predominantly inattentive (ADHD-I), predominantly hyperactive-impulsive (ADHD-HI), and combined (ADHD-C) (Magnus et al. 2025). The clinical presentation of ADHD changes over time; compared to children with the condition, adults are less likely to be hyperactive-impulsive and more likely to experience inattention. As overt hyperactive-impulsive symptoms abate with time, the remaining inattentive symptoms can be masked by compensatory strategies, making ADHD symptoms less noticeable in adults and contributing to missed or delayed diagnosis (Rivas-Vazquez et al. 2023). Notably, of the three ADHD subtypes, the predominantly inattentive subtype is the most common among adults (Ayano et al. 2023). Furthermore, while the existence of ADHD symptoms before the age of 12 is required to establish a diagnosis as per the Diagnostic and Statistical Manual of Mental Disorders, 5th Edition (DSM-5) (Magnus et al. 2025), observations in which some adults met the criteria for ADHD with no childhood history of the disorder suggest that there may be a variant of adult ADHD that is not a continuance of a childhood ADHD, challenging the theory of a merely neurodevelopmental origin (Caye et al. 2016; Moffitt et al. 2015). However, the validity of late-onset or adult (first)-onset ADHD remains highly debated (Hutt Vater et al. 2024; Waltereit et al. 2023).

Adults with ADHD often face educational and career underachievement. Their symptoms may also strain relationships with work colleagues, family members, friends, and partners. Moreover, ADHD is linked to higher rates of delinquency, imprisonment, and substance use disorder. Patients are also prone to driving-related dangers, such as collisions, speeding, and traffic violations. Overall, ADHD can interfere with almost every facet of adult life, particularly if it remains undiagnosed, untreated, or ineffectively managed (Ginsberg et al. 2014). Adult ADHD remains under-detected, despite the high prevalence, particular clinical presentation, and potential negative consequences leaving these individuals susceptible to impairments in many aspects of their lives and at risk for poor mental health outcomes, including stress, anxiety, and depression (Garcia-Argibay et al. 2024; Ogrodnik et al. 2023; Rivas-Vazquez et al. 2023).

In the absence of confirmed biomarkers, the diagnosis of ADHD is established clinically through a multi-step process that may include a detailed clinical interview considering personal and family history, self-report behavior rating scales identifying the presence of characteristic symptoms, observer reports from family and employers, and continuous performance tests evaluating impairments in attention and impulse control (Cabral et al. 2020). While these tools collectively provide comprehensive information for an ADHD diagnosis, the process is time-intensive and resource-demanding.

A screening tool is easy to administer, quicker, and less expensive than a full diagnostic assessment and thus can be offered to a wide population. While it cannot establish a definitive diagnosis of a condition, it can be considered a form of triage that helps flag individuals who are more likely to have a condition and require referral to in-depth evaluation (Newlove-Delgado and Ford 2020). To provide a screening tool for ADHD in adults in the general population, the World Health Organization (WHO) developed the Adult ADHD Self-Report Scale version 1.1 (ASRS) based on DSM-IV criteria (Kessler et al. 2005). There are two versions for the ASRS: a full version with 18 questions divided into two main symptom domains—9 assessing inattention, 9 assessing hyperactivity-impulsivity, and a shorter 6-question version (ASRS-6) (Kessler et al. 2005). The ASRS-6 proved superior sensitivity (68.7%v.56.3%), specificity (99.5%v.98.3%), total classification accuracy (97.9% v.96.2%), and Cohen's kappa (0.76 v.0.58) than the full version (Kessler et al. 2005). After being translated and validated in several languages (including Spanish, Turkish, Hungarian, and Swedish) in clinical and non-clinical samples, the ASRS-6 proved to be a psychometrically reliable and valid measure for adult ADHD screening (Daigre et al. 2009; Evren et al. 2016; Farcas et al. 2018; von Wallenberg Pachaly et al. 2024). The 6-item version encompasses four items retrieved from the inattention domain of the full ASRS and two items from the hyperactivity-impulsivity domain (Kessler et al. 2005).

Validation studies of the ASRS-6 have reported heterogeneous findings regarding its dimensionality, with some supporting a unidimensional structure (Green et al. 2018; Kessler et al. 2007; von Wallenberg Pachaly et al. 2024) and others favoring a two-factor model reflecting inattention and hyperactivity-impulsivity (Carlucci et al. 2017; Daigre et al. 2009; Farcas et al. 2018; Hesse 2013; Panagiotidi et al. 2023), depending on the population and context. Several studies across diverse samples have supported a two-factor structure. For example, in an adult

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Hungarian general community sample ($N = 567$), confirmatory factor analysis (CFA) supported a two-factor model with acceptable internal consistency (Total $\alpha = 0.72$; inattention $\alpha = 0.77$; hyperactivity $\alpha = 0.54$) (Farcas et al. 2018). Similarly, among Spanish outpatients with substance use disorders ($N = 80$), exploratory factor analysis with Varimax rotation identified two factors explaining 67.7% of the variance, with Cronbach's α of 0.82 for inattention and 0.52 for hyperactivity (Daigre et al. 2009). In a Turkish inpatient sample with alcohol use disorder ($N = 190$), principal component analysis (PCA) revealed two components accounting for 38.64% and 18.08% of the variance (Evren et al. 2016). Likewise, in a Canadian clinical sample of adults with eating disorders ($N = 500$), exploratory factor analysis followed by CFA confirmed a two-factor solution, with factor 1 (inattention) explaining 49.51% of the variance and factor 2 (hyperactivity-impulsivity) 19.98%, and internal consistency estimates of $\alpha = 0.83$ (inattention), $\alpha = 0.63$ (hyperactivity), and $\alpha = 0.78$ (total) (Carlucci et al. 2017). In contrast, other studies have supported a unidimensional structure. In a large U.S. adolescent community sample ($N = 2472$), one-factor CFA demonstrated good fit and internal consistency ($\alpha = 0.80$), with invariance across gender (Green et al. 2018). Similarly, in a Swedish psychiatric sample of young adults with ADHD and/or borderline personality disorder and/or bipolar disorder ($N = 151$), CFA supported a one-factor model with acceptable internal consistency ($\alpha = 0.74$) (von Wallenberg Pachaly et al. 2024). The variability in factorial validity and reliability across cultural and clinical contexts, underscores the importance of conducting context-specific psychometric evaluations when adapting screening tools such as the ASRS-6 for use in new linguistic and cultural settings.

Despite being used in a few studies involving Arabic-speaking populations such as Egypt (Abd El-Hay and El Sawy 2011; Khalil et al. 2016), Iraq (Ashor 2012), and Lebanon (Farchakh et al. 2022), to the best of our knowledge, neither the factor structure of the Arabic version of the ASRS-6 screener had not been tested, nor its psychometric properties have not been evaluated. The very few studies focusing on the topic showed that adult ADHD is prevalent in the Arab World, is highly comorbid with other mental health problems, and associated with significant impairment (Abd El-Hay and El Sawy 2011; Ashor 2012; Farchakh et al. 2022; Hayek et al. 2019; Khalil et al. 2016). Yet, it continues to be largely under-researched and undertreated throughout the Middle East and North Africa (Hayek et al. 2019). Providing a practical and psychometrically valid screening instrument for detecting potential adult ADHD cases from the general Arabic-speaking populations could promote research in this area, assist healthcare professionals in guiding referrals, and facilitate detection, which can reduce the burden of untreated ADHD-related impairments and improve quality of life. Furthermore, normalizing ADHD assessment in the general population could help reduce stigma and encourages people to seek help (Younes et al. 2024). To address the absence of a validated tool for ADHD screening in Arabic-speaking populations, this study aimed to: (1) examine the factor structure of the Arabic ASRS-6 using exploratory factor analysis followed by confirmatory factor analysis in a sample of Lebanese adults from the general population, and (2) assess the internal consistency, sex invariance, and concurrent validity of the screener. We hypothesized that the ASRS-6 would demonstrate acceptable reliability and validity, and invariance across gender.

2 | Methods

2.1 | Data Collection and Sampling Procedure

This cross-sectional study recruited 1059 participants between April and May 2025 using a mixed convenience and snowball sampling approach. The mean age of the participants was 28.52 years and 66.0% were females. Other descriptive statistics of the sample can be found in Table 1. Participants were recruited through a Google Forms survey distributed via social media platforms. The research team first shared the survey link with possible respondents, who were then encouraged to forward it within their networks to other eligible individuals. Prior to filling the questionnaire, the participants provided digitally their consent. Inclusion criteria included Lebanese nationality, age ≥ 18 years, ability to read Arabic, access to the internet, and willingness to participate. Individuals reporting a prior psychiatric diagnosis by a psychiatrist were excluded. The participation in the study was voluntary and unremunerated.

2.2 | Ethic Statement

The Lebanese International University's School of Pharmacy ethics committee granted this study ethics permission (2025ERC-013-LIUSOP). A written informed consent was obtained from each participant when completing the online form. All methods were performed in accordance with the relevant guidelines and regulations (in accordance with the Declaration of Helsinki).

2.3 | Minimal Sample Size Calculation

A minimum of 60 and 120 participants was required to conduct the exploratory and confirmatory factor analyses respectively, as per the recommendation of using 10 and 20 times (Comrey and Lee 2013) the number of scale variables to calculate the sample size for factor analysis, and considering the 6-question ASRS screener.

2.4 | Instruments

The questionnaire was prepared in Arabic, Lebanon's official language. It included socio-demographic data, the Adult ADHD Self-Report Scale Version 1.1 6-Question Screener, and three additional scales to assess concurrent validity. It commenced with an introduction describing the study's purpose and comprised an online consent form to secure voluntary participation and ensure confidentiality and anonymity. The socio-demographic segment retrieved information regarding age, sex, marital status, and education level. The four self-administered scales are described as follows:

Adult ADHD Self-Report Scale Version 1.1 6-Question Screener (ASRS-6) (Kessler et al. 2005): It is a subset of the WHO's 18-question Adult ADHD Self-Report Scale v1.1 based on DSM-IV criteria. It evaluates how often symptoms have arisen over the past 6 months on a 0–4 scale: never (0), rarely (1), sometimes (2), often (3), and very often (4). The total score, obtained by summing the points, ranges from 0 to 24, with a cut point of

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TABLE 1 | Sociodemographic characteristics of the sample ($N = 1059$).

Variable	*N* (%)
Sex	
Male	360 (34.0%)
Female	699 (66.0%)
Marital status	
Single	724 (68.4%)
Married	335 (31.6%)
Education	
Secondary or less	201 (19.0%)
University	858 (81.0%)
	Mean ± SD
Age (years)	28.52 ± 11.41
Household crowding index (person/room)	1.10 ± 0.60

Abbreviations: M, mean; N, frequency; SD, standard deviation.

14 or higher indicating a positive ADHD screening. Scores are categorized into four-stratum: low negative (0–9), high negative (10–13), low positive (14–17), and high positive (18–24) (Harvard Department of Health Care Policy 2024). The 6-question screener has been translated into a number of languages, including Arabic, using the standard WHO translation and back-translation protocol through an unrestricted educational grant from the Eli Lilly Company (HARVARD MEDICAL SCHOOL DEPARTMENT OF HEALTH CARE POLICY). The translations are available on the official website from which the Arabic version used in this study was retrieved (https://www.hcp.med.harvard.edu/ncs/asrs.php). The Cronbach's alpha in this study was 0.68.

Stress Numerical Rating Scale-11 (SNRS-11) (Karvounides et al. 2016): Validated in Arabic (Obeid, Hallit, et al. 2024a), the SNRS-11 is a single-item scale that asks participants to respond to the question, "On a scale of 0–10, with 0 being no stress and 10 being worst stress possible, what number best describes your level of stress right now?". Responses are classified as follows: no stress (0), mild stress (1–3), moderate stress (4–7), and severe stress (8–10).

Patient Health Questionnaire (PHQ-4) (Kroenke et al. 2009): Validated in Arabic (Obeid, Hemade, et al. 2024b), the PHQ-4 is an ultra-brief four-item scale designed to assess symptoms of anxiety and depression experienced over the past 2 weeks. Each of its two subscales (anxiety and depression) possesses two items. Answers are rated on a four-point scale: 0 ("Not at all"), 1 ("Several days"), 2 ("More than half the days"), and 3 ("Nearly every day"). Each subscale yields a score ranging from 0 to 6. Scores of $\geq$ 3 on the respective subscales indicate relevant symptoms of anxiety or depression. The Cronbach's alpha in this study was 0.75.

The *World Health Organization-Five Well-Being Index (WHO-5)* (World Health Organization 2024): Validated across six Arab countries (Fekih-Romdhane et al. 2023), the WHO-5 assesses subjective psychological well-being through five positively phrased items. Each item is rated on a five-point Likert scale ranging from 0 (At no time) to 5 (All of the time), based on how the respondent felt during the past 2-week period. The total score ranges from 0 to 25 and can be multiplied by 4 to yield a percentage score ranging from 0 (worst possible mental well-being) to 100 (best possible mental well-being). A percentage score below 50 (or a raw score below 13) has been suggested as a cut-off for poor mental well-being (Topp et al. 2015; World Health Organization (WHO) 2024). The Cronbach's alpha in this study was 0.90.

2.5 | Analytic Strategy

The total sample was randomly divided into two split-half subsamples using the SPSS option. The first split-half sample was analyzed using principal-axis exploratory factor analysis (EFA) within R software (The R Core Team 2026). Following the recommendations of (Hair et al. 2009), the suitability of the data for factor analysis was evaluated using the Kaiser-Meyer-Olkin (KMO) index of sampling adequacy and Bartlett's sphericity (expected to be significant). Principal-axis factoring was selected because it provides results comparable to maximum likelihood estimation while not requiring multivariate normality (Goretzko et al. 2021). Given the hypothesized presence of a single orthogonal factor, quartimax rotation was employed (Pedhazur and Schmelkin 2013).

We conducted a CFA via the *lavaan* (Rosseel 2012), *semTools* (Jorgensen et al. 2018) with R software on the second subsample. Multiple fit indices were calculated: root mean square error of approximation (RMSEA) ($\leq$ 0.08), standardized root mean square residual (SRMR) ($\leq$ 0.05), Tucker-Lewis Index (TLI) and Comparative Fit Index (CFI) ($\geq$ 0.90 for both) (Hu and Bentler 1999). Additionally, the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) were calculated for model comparisons. A difference $<$ 2 between AIC and BIC values indicated that both models were acceptable, whereas a difference $>$ 10 favored the model with the lower value (Burnham and Anderson 2004). Multivariate normality was not verified as shown by the Mardia skewness (167.59; $p < 0.001$) and kurtosis (8.67; $p < 0.001$) values, so parameter estimates

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were obtained using the robust maximum likelihood method. A multi-group CFA was conducted to examine measurement invariance of the ASRS-6 screener scores between sexes (Chen 2007) at the configural, metric, and scalar levels (Vandenberg and Lance 2000). ΔCFI $\leq$ 0.010 and ΔRMSEA $\leq$ 0.015 or ΔSRMR $\leq$ 0.010 supported the evidence of invariance (Swami et al. 2022). Comparison of ASRS-6 scores between sexes was done using the independent samples *t* test. Internal consistency was assessed using McDonald's ω and Cronbach's α, with values greater than 0.70 reflecting adequate internal consistency (Dunn et al. 2014). The association between the ASRS-6 scores and other scores was evaluated using the Pearson correlation test.

3 | Results

In total, 1059 individuals participated in this study, with a mean age of 28.52 years and 66.0% females. Other descriptive statistics of the sample can be found in Table 1.

The categorical distributions of ADHD screening and psychological measures, along with the corresponding mean scores, are presented in Table 2. Among the participants, 229 (21.6%) participants screened positive for ADHD. Moreover, 575 (54.3%) and 279 (26.3%) had moderate and severe stress, 488 (46.1%) had anxiety symptoms, 501 (47.3%) had depressive symptoms, and 381 (36.0%) had poor mental well-being.

3.1 | Exploratory Factor Analysis

Data from the first split-half subsample ($n = 528$) were subjected to principal-axis exploratory factor analysis. The Kaiser-Meyer-Olkin (KMO) measure of sampling adequacy was acceptable (KMO = 0.76), and Bartlett's test of sphericity was significant (χ^2 (15) = 418.23, $p < 0.001$), indicating that the data were suitable for factor analysis. Inspection of the scree plot (Figure 1) and results from parallel analysis suggested a two-factor solution. However, the second factor accounted for only a small proportion of variance (4%) and did not demonstrate a meaningful pattern of loadings, with most items loading primarily on the first factor. Therefore, the two-factor solution lacked interpretability.

Given that a one-factor solution was retained, rotation was not necessary as it does not affect single-factor models. All items demonstrated sampling adequacy (MSA > 0.70), supporting their retention.

Consequently, in line with theoretical expectations, a one-factor solution was retained. This model showed acceptable model fit χ^2 (9) = 30.29, $p < 0.001$; χ^2/df = 3.37; RMSEA = 0.067, 90%CI [0.042, 0.094]; TLI 0.91; RMSR = 0.040. Factor loadings ranged from 0.35 to 0.64, with all items exceeding the recommended threshold of 0.33. Communalities ranged from 0.12 to 0.41, indicating moderate item representation by the latent factor. Overall, the results supported a unidimensional structure of the screener.

A CFA was conducted on the second split-half subsample ($n = 531$) to test the one-factor model identified in the EFA. The initial one-factor model showed an acceptable but not optimal fit to the data (χ^2 (9) = 58.22, $p < 0.001$; CFI 0.928; TLI = 0.881; RMSEA = 0.102, 90% CI [0.08, 0.13]; SRMR = 0.051). All standardized factor loadings were statistically significant ($p < 0.001$), supporting the contribution of each item to the latent construct (Table 3).

Inspection of modification indices suggested a meaningful correlated residual between ADHD5 and ADHD6 (modification index = 17.48), indicating shared variance between these two items beyond the common latent factor. Given the conceptual similarity between these items, this covariance was added to the model. The revised one-factor CFA model including the residual covariance between items 5 and 6 demonstrated improved model fit and was retained as the final model (χ^2 (8) = 31.56; CFI = 0.966; TLI = 0.936; RMSEA = 0.075, 90% CI [0.048, 0.103]; SRMR = 0.037). All standardized factor loadings remained statistically significant ($p < 0.001$) and ranged from 0.32 to 0.71. The standardized residual covariance between items 5 and 6 was significant ($r = 0.21$, $p < 0.001$). These findings supported the retention of the six-item one-factor model with a correlated residual between ADHD5 and ADHD6 as the final factorial structure of the scale. Internal consistency

TABLE 2 | Descriptive statistics and categorical distribution of ADHD screening and psychological measures ($N = 1059$).

Domain	Measure	Mean ± SD	Category	*N* (%)
ADHD	ASRS-6 total score	10.63 ± 4.03	Negative screening (< 14)	830 (78.4%)
			Positive screening (≥ 14)	229 (21.6%)
Stress	SNRS-11 total score	5.68 ± 2.52	No stress (0)	53 (5.0%)
			Mild stress (1–3)	152 (14.4%)
			Moderate stress (4–7)	575 (54.3%)
			Severe stress (8–10)	279 (26.3%)
Anxiety	PHQ-4 anxiety subscale	2.46 ± 1.73	Irrelevant symptoms (< 3)	571 (53.9%)
			Relevant symptoms (≥ 3)	488 (46.1%)
Depression	PHQ-4 depression subscale	2.47 ± 1.61	Irrelevant symptoms (< 3)	558 (52.7%)
			Relevant symptoms (≥ 3)	501 (47.3%)
Well-being	WHO-5 total score	14.10 ± 5.56	Poor well-being (< 13)	381 (36.0%)
			Good well-being (≥ 13)	678 (64.0%)

Abbreviations: ASRS-6, Adult ADHD Self-Report Scale Version 1.1 6-Question Screener; M, mean; N, frequency; PHQ-4, Four-Item Patient Health Questionnaire for Anxiety and Depression; SD, standard deviation; SNRS-11, Stress Numerical Rating Scale-11; WHO-5, World Health Organization-Five Well-Being Index.

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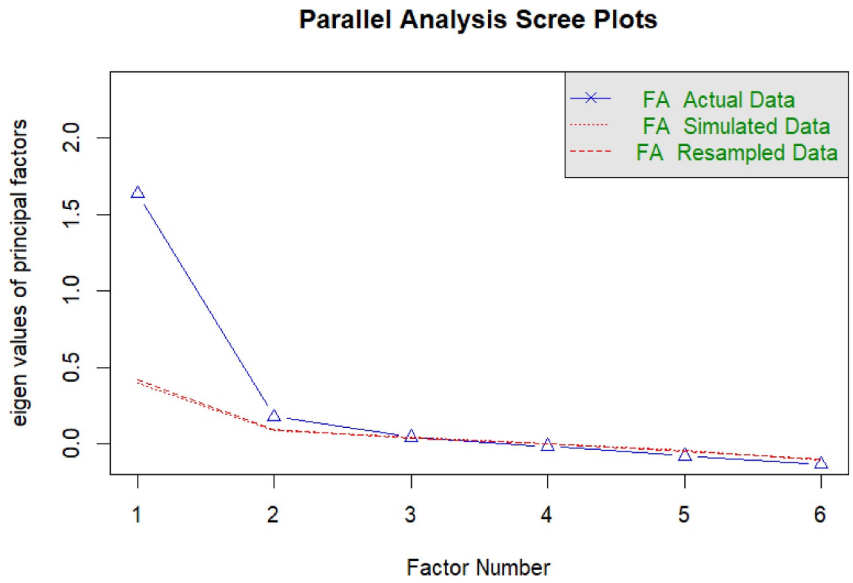


FIGURE 1 | Scree plot deriving from the exploratory factor analysis.

TABLE 3 | Standardized loading factors deriving from the exploratory (EFA) and confirmatory (CFA) factor analyses of the 6-item model.

Item	EFA	CFA (6-item model)	CFA (6-item model with correlation between residuals of items 5 and 6)
ADHD1	0.49	0.58	0.58
ADHD2	0.64	0.69	0.71
ADHD3	0.47	0.57	0.58
ADHD4	0.62	0.55	0.56
ADHD5	0.52	0.51	0.46
ADHD6	0.35	0.39	0.32

was borderline, with Cronbach's alpha = 0.68 and McDonald's omega = 0.69.

3.2 | Sensitivity Analysis: Five-Item Model Excluding ADHD6

Given that ADHD6 showed a factor loading < 0.4 in both the EFA and CFA, a sensitivity analysis excluding this item was conducted to examine whether a five-item solution would improve model fit.

In the EFA performed on the first subsample ($n = 528$), the five-item model showed slightly poorer factorial adequacy compared to the original six-item model. The Kaiser-Meyer-Olkin value decreased from 0.76 to 0.74, and Bartlett's test of sphericity was significant ($\chi^2(10) = 367.35$, $p < 0.001$), indicating that the data were suitable for factor analysis. Inspection of the scree plot and parallel analysis again suggested a two-factor solution; however, the first factor was clearly dominant and the second factor lacked theoretical interpretability. Therefore, a one-factor solution was retained.

The one-factor model showed slightly poorer fit compared to the original six-item model ($\chi^2(5) = 25.47$, $p < 0.001$; $\chi^2/df = 5.09$; TLI = 0.89; RMSEA = 0.088, 90% CI [0.056, 0.123]; SRMR = 0.05; BIC = −5.88). Factor loadings ranged from 0.49 to 0.65, with all items exceeding the recommended threshold of 0.33. Overall, although all remaining items loaded adequately, the factorial structure was less satisfactory than that observed for the six-item model.

TABLE 4 | Standardized loading factors deriving from the exploratory (EFA) and confirmatory (CFA) factor analyses of the 5-item model (after removal of item 6).

Item	EFA	CFA
ADHD1	0.49	0.57
ADHD2	0.65	0.72
ADHD3	0.49	0.58
ADHD4	0.60	0.55
ADHD5	0.50	0.46

A CFA was subsequently conducted on the second split-half subsample ($n = 531$) to test the one-factor five-item model. The model showed slightly improved fit indices compared to the initial six-item CFA model ($\chi^2(5) = 28.78$, $p < 0.001$; CFI = 0.96; TLI = 0.92; RMSEA = 0.095, 90% CI [0.063, 0.130]; SRMR = 0.041). All standardized factor loadings were statistically significant ($p < 0.001$) (Table 4).

Internal consistency remained borderline, with Cronbach's alpha = 0.68 and McDonald's Omega = 0.68. However,

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improvements in CFA fit were only marginal and did not meaningfully improve reliability or convergent validity. In contrast, the poorer EFA fit and reduced conceptual coverage argued against removing item 6. The six-item model also demonstrated a lower BIC value compared to the five-item model, further supporting the retention of the original factorial structure. Therefore, item 6 was retained to preserve the factorial integrity and conceptual completeness of the scale. Overall, the six-item one-factor model with a correlated residual between ADHD5 and ADHD6 was retained as the final and most theoretically appropriate structure of the scale.

3.3 | Sex Invariance

Full configural and metric invariances were supported in the one-factor model, with fit indices falling within the acceptable ranges (Table 5). However, full scalar invariance was not supported, as indicated by substantial declines in model fit (ΔCFI = 0.113 and ΔRMSEA = 0.066). To try to achieve some level of scalar invariance, partial invariance was evaluated by allowing the intercepts of non-invariant items to vary while constraining the rest. Acceptable model-fit indices were then achieved (ΔCFI = 0.057 and ΔRMSEA = 0.034), confirming partial scalar invariance and indicating that comparisons of mean scores across sexes would be acceptable using this instrument. A significantly higher ASRS-6 mean scores was found in females versus males (10.85 $\pm$ 3.98 vs 10.22 $\pm$ 4.09; t (1057) = -2.41, p = 0.009, Cohen's d = 0.157). In terms of percentages, no significant difference was found between the females who screened positive for ADHD compared to males (n = 163 (23.3%) versus n = 66 (18.3%), X^2 = 3.49, p = 0.070, Effect size = 0.062). Moreover, a significantly higher ASRS-6 mean score was found in single participants compared to married ones (10.84 $\pm$ 3.95 vs 10.19 $\pm$ 4.16; t (1057) = 2.46, p = 0.014, Cohen's d = 0.163). However, no significant difference was found between participants with a secondary level of education or less compared to those with a university education level in terms of ASRS-6 scores (10.20 $\pm$ 4.32 vs 10.73 $\pm$ 3.95; t (1057) = -1.68, p = 0.094, Cohen's d = 0.131).

3.4 | Concurrent Validity

Higher ASRS-6 scores were significantly associated with higher depression (r = 0.33; $p < 0.001$), anxiety (r = 0.33; $p < 0.001$), stress (r = 0.30; $p < 0.001$), and lower well-being scores ($r = -0.38$; $p < 0.001$) (Table 6).

4 | Discussion

This study aimed to assess the factor structure and other psychometric properties of the Arabic translation of the Adult ADHD Self-Report Scale v1.1 6-Question screener among Arabic-speaking adults from the general population of Lebanon. The results may suggest a one factorial structure based on EFA and CFA, with adequate internal consistency, partial sex invariance, and evidence supporting its concurrent validity. Overall, the findings could provide initial evidence in favor of utilizing the Arabic version of this instrument for adult ADHD screening.

Of the 1059 participants, 229 (21.6%) screened positive for ADHD using the ASRS-6. Comparisons are not possible because, as far as we are, no prior epidemiological studies in Lebanon have estimated the prevalence of adult ADHD (Assaf et al. 2024). However, a positive result does not confirm ADHD despite the screener strong concordance with clinician diagnoses (Kessler et al. 2007). Individuals screened positive should undergo a psychiatric evaluation; it is essential to emphasize that a screening instrument only serves as a guiding tool and cannot substitute a comprehensive clinical assessment to establish a diagnosis.

In terms of factor structure, the EFA results supported a unifactorial structure for the screener which was confirmed by CFA. Similar to our findings, some studies reported a unidimensional structure, including a study on a sample of subscribers to a large health plan in the US (Kessler et al. 2007), a study on a US adolescent community sample (Green et al. 2018), and another on a Swedish patient sample with ADHD and/or borderline personality disorder, and/or bipolar disorder (von Wallenberg Pachaly et al. 2024). Kessler et al. explained that the absence of a second factor in the ASRS-6 could be a joint function of the limited number of items and the strong intercorrelations between inattention and hyperactivity-impulsivity symptoms; even though the symptoms of ADHD are known to be two-dimensional, they can be captured by a single underlying dimension whereas ADHD as a disorder includes both types of symptoms (Kessler et al. 2007). However, a two-factor structure has often been reported in prior studies

TABLE 6 | Correlation matrix.

	1	2	3	4
1. ADHD	1			
2. Depression	0.33***	1		
3. Anxiety	0.33***	0.64***	1	
4. Stress	0.30***	0.41***	0.47***	1
5. Well-being	−0.38***	−0.46***	−0.47***	−0.48***

Note: Numbers refer to Pearson correlation coefficients (r).
***$p < 0.001$.

TABLE 5 | Measurement Invariance of the Adult ADHD Self-Report Scale v1.1 6-Question Screener Across Sexes.

Model	CFI	RMSEA	SRMR	Model comparison	ΔCFI	ΔRMSEA	ΔSRMR
One-factor model							
Configural	0.994	0.025	0.032				
Metric	0.991	0.027	0.037	Configural versus metric	0.003	0.002	0.005
Scalar	0.934	0.061	0.043	Metric versus scalar	0.057	0.034	0.006

Abbreviations: CFI, Comparative fit index; RMSEA, root mean square error of approximation; SRMR, Standardized root mean square residual.

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conducted with various populations, including Hungarian adults from the general community (Farcas et al. 2018), Spanish patients with substance use disorders (SUD) (Daigre et al. 2009), a Danish sample of university students, and another sample of patients with SUD (Hesse 2013), a Turkish sample of inpatients with alcohol use disorder (Evren et al. 2016), and a Canadian clinical sample with eating disorders (Carlucci et al. 2017). The two-factor model aligns with the theoretical and diagnostic frameworks of ADHD, which conceptualize the disorder as two dimensions of inattention and hyperactivity-impulsivity symptoms that are used to define its three subtypes: ADHD-I, ADHD-HI, and ADHD-C (Administration and Substance Abuse and Mental Health Services 2016). These discrepancies in factor structures may be partly explained by differences in sample characteristics and study settings. Studies supporting a two-factor structure have often been conducted in clinical populations e.g., substance use or eating disorder samples (Carlucci et al. 2017; Daigre et al. 2009; Evren et al. 2016; Hesse 2013), where symptom differentiation may be more pronounced. In contrast, unidimensional solutions are more frequently observed in large community-based samples (Green et al. 2018; Kessler et al. 2007), where symptom severity is lower and dimensions may be less distinct (Niina et al. 2022).

The internal consistency of the ASRS-6 was adequate for the total score ($\omega = 0.68$, $\alpha = 0.68$) supporting its reliability as a unidimensional screening measure. This estimate is comparable to those reported in previous studies (e.g., $\alpha = 0.72$ in (Farcas et al. 2018); $\alpha = 0.78$ in (Carlucci et al. 2017).), and is in line with expectations for brief scales, as reliability coefficients are sensitive to the number of items (Tavakol and Dennick 2011).

Sex invariance examines whether an instrument is interpreted equivalently by males and females. For both models of the ASRS-6, full configural and metric invariance were established. However, partial invariance was established at the scalar level, suggesting that although some items function differently across genders, the overall latent construct remains comparable, allowing comparisons of latent mean differences (Putnick and Bornstein 2016). In the current study, ADHD positive screening rates were comparable between males and females. In a review by Faheem et al. (2022) on sex-based differences in the prevalence of ADHD in adults, 8 out of 20 studies reported an equal prevalence among males and females, with one study indicating a higher prevalence in females (male-to-female ratio 2:3). In contrast, the remaining 11 studies found a higher prevalence in males, with male-to-female ratios ranging from 1.5:1 to 3:1 (Faheem et al. 2022). Despite increasing research on sex differences in ADHD, little is known about differences in its prevalence by sex in adults, and there remains a dearth of reviews limiting the ability to draw firm conclusions.

Regarding the assessment of concurrent validity, it was supported by significant positive correlations between ASRS-6 scores and measures of depression, anxiety, and stress, as well as negative correlations with mental well-being scores. These findings showed the expected patterns of correlations between ASRS-6 scores and the study variables and are, therefore, consistent with existing literature. Within the internalizing domain, anxiety disorders are the most commonly comorbid with ADHD and are associated with worse clinical presentation (e.g., lower level of education, suicidal attempt, higher rate of hospitalization) (Quenneville et al. 2022; Ryan et al. 2023). Anxiety may arise as a consequence of ADHD due to apprehension stemming from past failures. Additionally, ADHD and anxiety disorders share common neurobiological dysfunctions (D'Agati et al. 2019; Ryan et al. 2023). ADHD symptoms could elicit stressful environments, such as failures in work and relationships, which in turn increase the risk of depression (Knouse et al. 2013; Panevska et al. 2015). Shared genetic factors can also account for the co-occurrence of ADHD and depression (Powell et al. 2021; Riglin et al. 2021).

4.1 | Limitations

We advise interpreting the results cautiously, taking into thought the following limitations, until these results are replicated. First, a non-random sample may have occurred due to the snowball sampling technique, which may have limited the sample's representativeness to the wider Lebanese adult population. Second, while self-report measures speed up data collection, they are vulnerable to interpretation and response bias. Third, the analysis was limited to finding correlations between the measured variables because this study was conducted at a single time, restricting its ability to establish causal relationships. Additionally, our study did not employ clinical assessments to confirm the presence of adult ADHD, given that it was an online, anonymous, cross-sectional survey. While it would have been ideal, conducting clinician assessments in a large community sample would have been costly. Nevertheless, we recognize the importance of validating the Arabic version of the ASRS-6 against a gold-standard ADHD diagnosis for future research contributions. In addition, other aspects of validity were not assessed in this study. Convergent validity with ADHD-specific instruments, known-groups validity, and diagnostic accuracy (e.g., ROC analysis and optimal cut-off determination) could not be evaluated as the collected data in this study did not include appropriate comparator measures and clinical diagnostic data required for such analyses. Furthermore, test–retest reliability was not examined. Future research should address these limitations to strengthen the psychometric evaluation of the ASRS-6 and enhance confidence in its temporal stability. Finally, the validation of the Arabic version was limited to a sample of Lebanese adults, which may not represent the entire Arab world, necessitating further studies to confirm its validity across various Arab countries.

4.2 | Practical Implications

Despite these limitations and the need for further research, this study may provide initial evidence for the reliability and validity of the Arabic version of the ASRS-6. Such a screener, by facilitating data collection on adult ADHD prevalence and correlates, could increase epidemiological research within Arabic-speaking nations. Moreover, it represents a valuable tool for clinical outreach and case-finding initiatives in academic settings, workplaces, and primary care due to its ease of self-administration and speed (in less than 2 minutes) (Kessler et al. 2007). Widespread screening may facilitate the detection of ADHD cases, enabling timely interventions that reduce untreated ADHD-related impairments, ultimately enhancing productivity and quality of life. In addition, the screener may

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contribute to raised awareness and lesson misconceptions about ADHD, particularly within Arabic-speaking communities where mental health issues are often stigmatized (Younes et al. 2024). Such efforts may foster a more informed and accepting perspective on ADHD, encouraging individuals to seek professional assistance.

5 | Conclusion

According to the preliminary findings, the Arabic version of the ASRS-6 seems valid for screening adult ADHD in the general population. However, to confirm its validity, assess additional psychometric properties, and validate it across other Arab nations to improve its applicability, further research is needed. The availability of such a brief, free, and easy-to-administer screener is a significant step toward advancing ADHD epidemiological research and case-finding initiatives in Arabic-speaking populations. A comprehensive clinical evaluation should be performed on those who screen positive for ADHD with this tool in order to confirm the diagnosis and rule out comorbid conditions.

Author Contributions

Melyssa Assaf: writing – review and editing, validation, visualization, writing – original draft. **Nelly Kheir:** formal analysis, visualization, validation, writing – review and editing, writing – original draft. **Rabih Hallit:** writing – review and editing, visualization, methodology, validation. **Diana Malaeb:** writing – review and editing, validation, visualization, investigation, methodology, project administration. **Kamel Jebreen:** writing – review and editing, formal analysis, visualization, validation. **Fouad Sakr:** writing – review and editing, validation, visualization, investigation. **Mariam Dabbous:** writing – review and editing, visualization, validation, investigation. **Sahar Obeid:** conceptualization, project administration, writing – review and editing, visualization, validation, methodology, supervision. **Feten Fekih-Romdhane:** conceptualization, writing – review and editing, methodology, validation, visualization, supervision, project administration. **Souheil Hallit:** data curation, validation, formal analysis, project administration, supervision, writing – original draft, writing – review and editing, visualization, methodology, conceptualization.

Acknowledgments

The authors would like to thank all participants.

Funding

The authors have nothing to report.

Consent

The authors have nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author but restrictions apply to the availability of these data, which were used under license for the current study, and so are not publicly available. Data are however available from the authors upon reasonable request and with permission of the ethics committee.

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